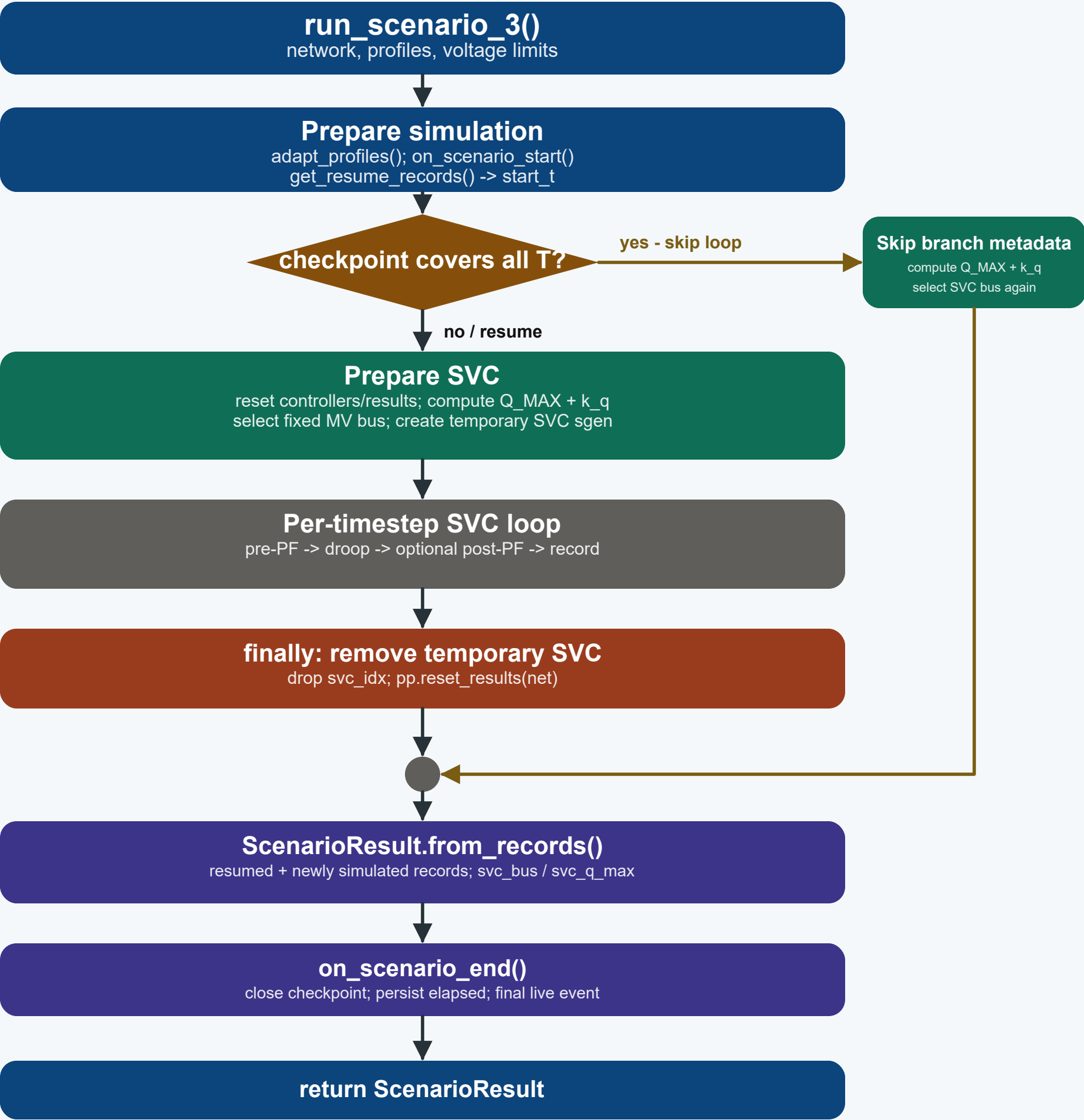




Fixed SVC bus selection

Eligible buses: in-service buses at the dominant MV level, excluding ext_grid slack.

Primary: deepcopy(net) -> apply_overvoltage_stress() -> runpp().

Stress state: sgen P = 0.90*sn_mva; load P and Q = 0.20*nominal; sgen Q=0.

Primary result: choose the eligible MV bus with minimum vm_pu in that stress snapshot.

Fallback 1: scan up to 7*24*6 profile steps; choose worst violated MV bus at first violation.

Fallback 2: use mv_buses[-1].

SVC parameter / lifecycle semantics

$Q_MAX = 0.20 * \text{sum}(\text{net.trafo.sn_mva})$.

$k_q = Q_MAX / (0.04 - 0.01) = Q_MAX / 0.03$.

The SVC is created only on the simulation branch, after bus selection.

Its sgen is removed in finally, so cleanup also runs if an uncaught loop exception escapes.

Checkpoint and failure semantics

on_scenario_start() runs before get_resume_records().

A complete checkpoint skips SVC creation and the timestep loop, but still recomputes Q_MAX and bus selection.

Partial resume creates a new SVC and continues at start_t using resumed_records as the prefix.

Selection / setup exceptions occur before the loop try/finally and therefore propagate without SVC cleanup.

Loop exceptions after SVC creation execute the finally cleanup before propagating.

on_scenario_end() closes the checkpoint handle and appends the scenario_complete live event.